Experiment-14

Write A Python Program To Implement Alpha And Beta Pruning Algorithm For Gaming

Aim

To write and execute a Python program to implement the Alpha-Beta Pruning algorithm to find the optimal value in a game tree while reducing unnecessary searches.

Algorithm

1. Start the program.
2. Define the leaf node values of the game tree.
3. Initialize alpha (\(\alpha\)) to negative infinity.
4. Initialize beta (\(\beta\)) to positive infinity.
5. Begin the search from the root node.
6. If the node is a leaf node, return its value.
7. If it is a maximizing node, select the maximum value.
8. Update alpha with the best value found.
9. If it is a minimizing node, select the minimum value.
10. Update beta with the best value found.
11. If alpha is greater than or equal to beta, prune the remaining branches.
12. Return and display the optimal value.

Code

```
# Alpha-Beta Pruning Algorithm

def alpha_beta(depth, node, maximizing, values,
               alpha, beta, max_depth):

    if depth == max_depth:
        return values[node]

    if maximizing:
        best = float('-inf')

        for i in range(2):
            value = alpha_beta(
                depth + 1,
                node * 2 + i,
                False,
                values,
                alpha,
                beta,
                max_depth
            )

            best = max(best, value)
            alpha = max(alpha, best)

            if beta <= alpha:
                break

        return best
```

```
    else:
        best = float('inf')

        for i in range(2):
            value = alpha_beta(
                depth + 1,
                node * 2 + i,
                True,
                values,
                alpha,
                beta,
                max_depth
            )

            best = min(best, value)
            beta = min(beta, best)

            if beta <= alpha:
                break

        return best

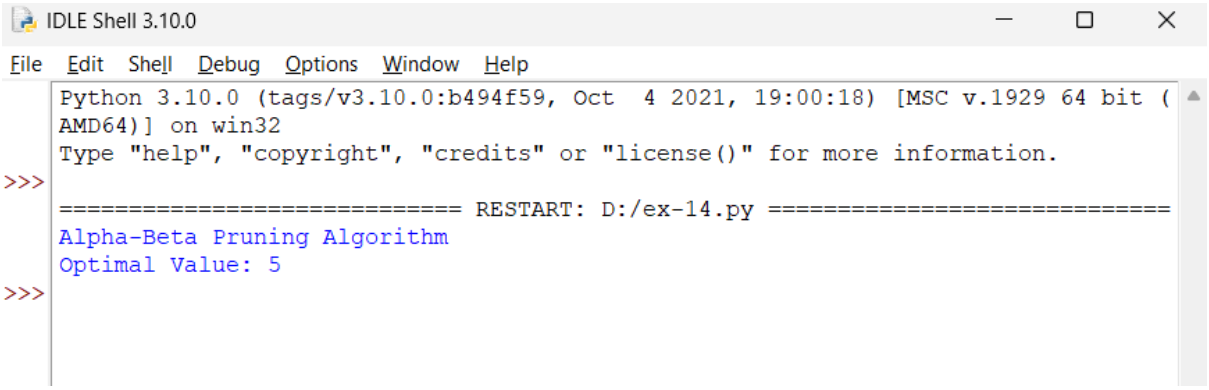
# Leaf node values
values = [3, 5, 6, 9, 1, 2, 0, -1]

max_depth = 3

result = alpha_beta(
    0, 0, True, values,
    float('-inf'),
    float('inf'),
    max_depth
)

print("Alpha-Beta Pruning Algorithm")
print("Optimal Value:", result)
```

Output



Result

The Alpha-Beta Pruning algorithm was successfully implemented. The program evaluates the game tree, prunes unnecessary branches, and obtains the optimal value of 5.